ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 7

NATURAL SCIENCES

PRACTICE EXAM 1

MARKS: 100 **TIME: 2 hours**

Covers all four Grade 7 strands: Life and Living (biosphere, biodiversity, reproduction), Matter and Materials (properties, mixtures, acids and bases, periodic table), Energy and Change (heat transfer, energy transfer, electricity supply) and Planet Earth and Beyond (Sun, Earth and Moon, astronomy).

Written by Academic Central for practice, following the CAPS curriculum for Grade 7. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B. Answer ALL the questions.
2. Number your answers exactly as the questions are numbered.
3. Read each question carefully and answer only what is asked.
4. Write in full sentences where explanations are asked for.
5. Diagrams are NOT necessarily drawn to scale.

SECTION A

QUESTION 1: MULTIPLE CHOICE

Choose the correct answer. Write only the letter (A–D) next to the question number, e.g. 1.11 B.

1.1 The part of a flower that produces pollen is the …

A stigma. B ovary. C sepal. D anther.

1.2 Which group of animals has feathers?

A Reptiles B Birds C Amphibians D Mammals

1.3 Which method is best for separating sand from water?

A Evaporation B Magnetism C Chromatography D Filtration

1.4 An acid turns blue litmus paper …

A blue. B red. C green. D colourless.

1.5 The chemical symbol for sodium is …

A S B So C Na D N

1.6 The transfer of heat through a solid, from particle to particle, is called …

A conduction. B convection. C radiation. D insulation.

1.7 A ball resting at the top of a hill has mainly …

A kinetic energy. B gravitational potential energy.
C sound energy. D electrical energy.

1.8 Which of these is a renewable energy source?

A Coal B Natural gas C Petrol D Wind

1.9 The seasons on Earth are caused by …

A the tilt of the Earth's axis as it moves around the Sun. B the changing distance between the Earth and the Sun.
C the phases of the Moon. D clouds blocking the Sun.

1.10 The scientist who suggested that the Sun, not the Earth, is at the centre of the solar system was …

A Ptolemy. B Newton. C Copernicus. D Aristotle.

(10)

QUESTION 2: SCIENTIFIC TERMS

Give the correct scientific term for each description. Write only the term next to the question number.

2.1 All the parts of the Earth where life is found

2.2 The transfer of pollen from an anther to a stigma

2.3 An animal without a backbone

2.4 Differences between individuals of the same species

2.5 A substance that changes colour to show whether a solution is acidic or basic

2.6 A pure substance made of only one kind of atom

2.7 A material that does not allow heat to pass through it easily

2.8 The energy that a moving object has

2.9 The different shapes of the lit part of the Moon that we see from Earth

2.10 The path that the Earth follows around the Sun

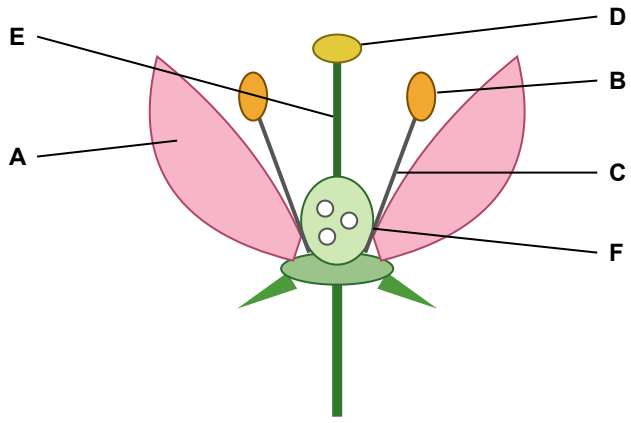
(10)

TOTAL SECTION A: [20]

SECTION B

QUESTION 3: LIFE AND LIVING

3.1 Study the diagram of a flower cut in half.



3.1.1 Identify the parts labelled A, B, D and F. (4)

3.1.2 Which TWO letters together make up the stamen? (2)

3.1.3 State ONE function of part F. (1)

3.2 Give TWO features of flowers that are pollinated by insects. (2)

3.3 Explain the difference between *pollination* and *fertilisation*. (2)

3.4 Name the way each of these seeds is dispersed: (a) blackjack seeds with tiny hooks; (b) dandelion seeds with fluffy "parachutes"; (c) coconuts that float. (3)

3.5 Classify each vertebrate into its class: (a) frog; (b) bat; (c) snake. (3)

3.6 (a) Name TWO changes that take place in BOTH boys and girls during puberty. (b) In which part of the female reproductive system does fertilisation normally take place? (3)

[20]

QUESTION 4: MATTER AND MATERIALS

4.1 Lindiwe is given a mixture of salt, sand and iron filings. Describe, step by step, how she can separate all three substances. (5)

4.2 A learner puts a dot of black koki ink on filter paper and dips the bottom of the paper in water. After a while the ink spreads into blue, yellow and pink bands. (a) Name this separation method. (b) What does the result tell you about black ink? (2)

4.3 Litmus paper was used to test three solutions.

Solution	Blue litmus	Red litmus
A: lemon juice	turns red	stays red
B: soap solution	stays blue	turns blue
C: sugar water	stays blue	stays red

4.3.1 Classify each solution as an acid, a base or neutral. (3)

4.3.2 Give ONE safety rule when working with acids in the laboratory. (1)

4.4 Answer the questions about the periodic table.

4.4.1 Write down the chemical symbols for oxygen, iron and calcium. (3)

4.4.2 Name the elements with the symbols K and Cu. (2)

4.4.3 On which side of the periodic table are most metals found? (1)

4.4.4 Give TWO properties of metals. (2)

4.5 Explain why a cooking pot is made of metal but its handle is made of plastic or wood. (1)

[20]

QUESTION 5: ENERGY AND CHANGE

5.1 Name the method of heat transfer in each case: (a) your hands get warm when you sit near a fire; (b) a metal spoon in hot tea becomes hot; (c) water in a pot moves round and round as it is heated. (3)

5.2 Explain why the heating element of a kettle is placed at the bottom and not at the top. (3)

5.3 (a) Give TWO ways a family can insulate their house to keep it warm in winter. (b) Explain why a ceiling helps to keep heat inside a house. (3)

5.4 (a) Write down the energy chain for a torch that is switched on, starting with the energy stored in the battery. (b) A light bulb gives off light and heat. Which energy is useful and which is wasted? (5)

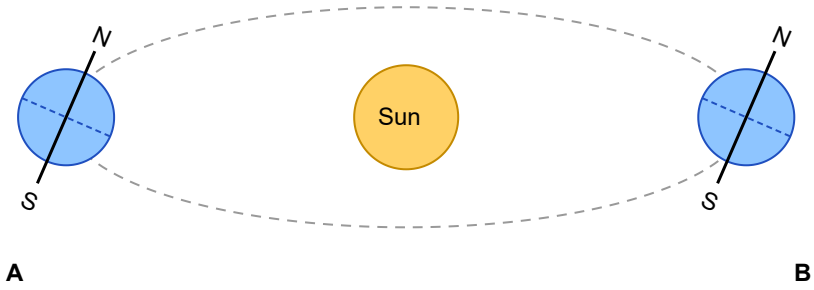
5.5 Describe how electricity is produced in a coal-fired power station, from burning the coal to the electricity leaving the station. (4)

5.6 Give TWO disadvantages of using coal to produce electricity. (2)

[20]

QUESTION 6: PLANET EARTH AND BEYOND

6.1 The diagram shows the Earth at two positions, A and B, as it moves around the Sun.



A **B**

The Earth's axis is tilted at 23,5° and always points in the same direction in space.

6.1.1 Which season is it in South Africa when the Earth is at position A? (1)

6.1.2 Explain your answer to 6.1.1. (2)

6.2 What does each of these movements cause? (a) The Earth spins on its axis once every 24 hours. (b) The Earth moves once around the Sun in about 365¼ days. (2)

6.3 (a) Why can we see the Moon at night, even though it does not make its own light? (b) Name the FOUR main phases of the Moon in order, starting with new moon. (c) About how long does it take to go through all the phases once? (6)

6.4 What causes tides in the oceans? (2)

6.5 Explain what happens during a solar eclipse. (2)

6.6 Compare the model of the universe described by Ptolemy with the model described by Copernicus. (2)

6.7 How did Galileo's observations with a telescope support the model of Copernicus? (2)

6.8 Name ONE large telescope in South Africa. (1)

[20]

TOTAL SECTION B: [80]

TOTAL: 100

END OF QUESTION PAPER